
DISCRETE STRUCTURES I

Sets

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1 Introduction

In this section, we study the fundamental discrete structure on which all other discrete structures are built, namely, the set. Sets are used to group objects together. Often, but not always, the objects in a set have similar properties. For instance, all the students who are currently enrolled in your school make up a set. Likewise, all the students currently taking a course in discrete mathematics at any school make up a set. In addition, those students enrolled in your school who are taking a course in discrete mathematics form a set that can be obtained by taking the elements common to the first two collections. The language of sets is a means to study such collections in an organized fashion. We now provide a definition of a set. This definition is an intuitive definition, which is not part of a formal theory of sets.

Definition 1. A set is an unordered collection of objects, called elements or members of the set. A set is said to contain its elements. We write $a \in A$ to denote that a is an element of the set A . The notation $a \notin A$ denotes that a is not an element of the set A .

Example 1 : The set V of all vowels in the English alphabet can be written as $V = \{a, e, i, o, u\}$.

Example 2 : The set O of odd positive integers less than 10 can be expressed by $O = \{1, 3, 5, 7, 9\}$.

Example 2 : The set of positive integers less than 100 can be denoted by $\{1, 2, 3, \dots, 99\}$.

Another way to describe a set is to use set builder notation. We characterize all those elements in the set by stating the property or properties they must have to be members. For instance, the set O of all odd positive integers less than 10 can be written as :

$$O = \{x \mid x \text{ is an odd positive integer less than } 10\}$$

Examples :

- $O = \{x \in \mathbb{Z}^+ \mid x \text{ is odd and } x < 10\}$.
- $\mathbb{Q} = \{x \in \mathbb{R} \mid x = \frac{p}{q} \text{ for some integers } p \text{ and } q\}$.

Usual Sets :

- $\mathbb{N} = \{0, 1, 2, 3, \dots\}$, the set of natural numbers.

- $\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$, the set of integers.
- $\mathbb{Z}^+ = \{1, 2, \dots\}$, the set of positive integers.
- $\mathbb{Q} = \{\frac{p}{q} \mid p \in \mathbb{Z}, q \in \mathbb{Z}, \text{ and } q \neq 0\}$, the set of rational numbers.

Definition 2. Two sets are equal if and only if they have the same elements. Therefore, if A and B are sets, then A and B are equal if and only if $\forall x (x \in A \leftrightarrow x \in B)$. We write $A = B$ if A and B are equal sets.

Example :

The sets $\{1, 3, 5\}$ and $\{3, 5, 1\}$ are equal, because they have the same elements. Note that the order in which the elements of a set are listed does not matter. Note also that it does not matter if an element of a set is listed more than once, so $\{1, 3, 3, 3, 5, 5, 5, 5\}$ is the same as the set $\{1, 3, 5\}$ because they have the same elements.

Remark :

There is a special set that has no elements. This set is called the empty set, or null set, and is denoted by ϕ . The empty set can also be denoted by $\{\}$ (that is, we represent the empty set with a pair of braces that encloses all the elements in this set). Often, a set of elements with certain properties turns out to be the null set. For instance, the set of all positive integers that are greater than their squares is the null set.

Venn Diagrams : Sets can be represented graphically using Venn diagrams. In Venn diagrams the universal set U , which contains all the objects under consideration, is represented by a rectangle. (Note that the universal set varies depending on which objects are of interest.) Inside this rectangle, circles or other geometrical figures are used to represent sets. Sometimes points are used to represent the particular elements of the set. Venn diagrams are often used to indicate the relationships between sets. Example :

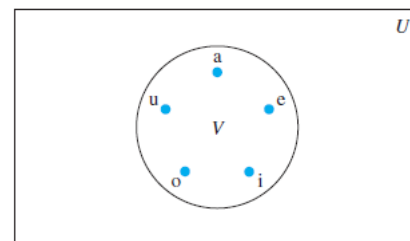


FIGURE 1 Venn Diagram for the Set of Vowels.

Definition 3. The set A is a subset of B if and only if every element of A is also an element of B . We use the notation $A \subseteq B$ to indicate that A is a subset of the set B .

We see that $A \subseteq B$ if and only if the quantification $\forall x (x \in A \rightarrow x \in B)$ is True.

Remark.

- To show that $A \subseteq B$, show that if $x \in A$ then x also belongs to B .
- To show that $A \not\subseteq B$, find a single $x \in A$ such that $x \notin B$.

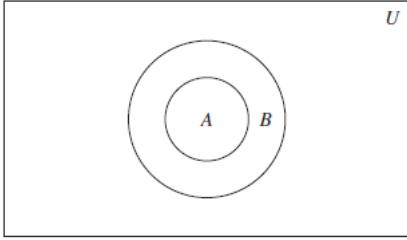


FIGURE 2 Venn Diagram Showing that A Is a Subset of B .

Theorem 1. For every set S :

1. $\phi \subseteq S$.
2. $S \subseteq S$.

Remark : To show that two sets A and B are equal, show that $A \subseteq B$ and $B \subseteq A$.

Definition 4. Let S be a set. If there are exactly n distinct elements in S where n is a nonnegative integer, we say that S is a finite set and that n is the cardinality of S . The cardinality of S is denoted by $|S|$.

Examples :

1. Let A be the set of odd positive integers less than 10. Then $|A| = 5$.
2. Let S be the set of letters in the English alphabet. Then $|S| = 26$.
3. $|\phi| = 0$.

Definition 5. A set is said to be infinite if it is not finite.

Remark : The set of positive integers is infinite.

Definition 6. Given a set S , the power set of S is the set of all subsets of the set S . The power set of S is denoted by $P(S)$.

Examples :

1. What is the power set of the set $\{0, 1, 2\}$?
The power set $P(\{0, 1, 2\})$ is the set of all subsets of $\{0, 1, 2\}$. Hence, $P(\{0, 1, 2\}) = \{\phi, \{0\}, \{1\}, \{2\}, \{0, 1\}, \{0, 2\}, \{1, 2\}, \{0, 1, 2\}\}$. Note that the empty set and the set itself are members of this set of subsets.
2. What is the power set of the empty set? What is the power set of the set $\{\phi\}$?
 $P(\phi) = \{\phi\}$.
The set $\{\phi\}$ has exactly two subsets, namely, ϕ and the set $\{\phi\}$ itself. Therefore, $P(\{\phi\}) = \{\phi, \{\phi\}\}$.
3. If a set has n elements, then its power set has 2^n elements.

The order of elements in a collection is often important. Because sets are unordered, a different structure is needed to represent ordered collections. This is provided by ordered n -tuples.

Definition 7. The ordered n -tuple $(a_1, a_2, \dots, a_n)$ is the ordered collection that has a_1 as its first element, a_2 as its second element, $\dots$, and a_n as its n th element.

We say that two ordered n -tuples are equal if and only if each corresponding pair of their elements is equal. In other words, $(a_1, a_2, \dots, a_n) = (b_1, b_2, \dots, b_n)$ if and only if $a_i = b_i$ for $i = 1, 2, \dots, n$. In particular, ordered 2-tuples are called ordered pairs. The ordered pairs (a, b) and (c, d) are equal if and only if $a = c$ and $b = d$. Note that (a, b) and (b, a) are not equal unless $a = b$.

Definition 8. Let A and B be sets. The Cartesian product of A and B , denoted by $A \times B$, is the set of all ordered pairs (a, b) , where $a \in A$ and $b \in B$. Hence, $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Example :

$A = \{1, 2\}$ and $B = \{a, b, c\}$

The Cartesian product $A \times B$ is : $A \times B = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\}$.

The Cartesian product $B \times A$ is : $B \times A = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$.

This is not equal to $A \times B$.

Definition 9. The Cartesian product of the sets $A_1, A_2, \dots, A_n$, denoted by $A_1 \times A_2 \times \dots \times A_n$, is the set of ordered n -tuples $(a_1, a_2, \dots, a_n)$, where a_i belongs to A_i for $i = 1, 2, \dots, n$. In other words, $A_1 \times A_2 \times \dots \times A_n = \{(a_1, a_2, \dots, a_n) \mid a_i \in A_i \text{ for } i = 1, 2, \dots, n\}$.

Example : Let $A = \{0, 1\}$, $B = \{1, 2\}$, and $C = \{0, 1, 2\}$.

The Cartesian product $A \times B \times C$ consists of all ordered triples (a, b, c) , where $a \in A$, $b \in B$, and $c \in C$. Hence : $A \times B \times C = \{(0, 1, 0), (0, 1, 1), (0, 1, 2), (0, 2, 0), (0, 2, 1), (0, 2, 2), (1, 1, 0), (1, 1, 1), (1, 1, 2), (1, 2, 0), (1, 2, 1), (1, 2, 2)\}$.

Definition 10. Given a predicate P , and a domain D , we define the truth set of P to be the set of elements $x \in D$ for which $P(x)$ is true.

The truth set of $P(x)$ is denoted by $\{x \in D \mid P(x)\}$.

Exercises : What are the truth sets of the predicates $P(x)$, $Q(x)$, and $R(x)$, where the domain is the set of integers and $P(x)$ is $|x| = 1$, $Q(x)$ is $x^2 = 2$ and $R(x)$ is $|x| = x$.

2 Set Operations

Two, or more, sets can be combined in many different ways.

Definition 11. Let A and B be sets. The union of the sets A and B , denoted by $A \cup B$, is the set that contains those elements that are either in A or in B , or in both.

$$A \cup B = \{x \mid x \in A \vee x \in B\}$$

. The union of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 2, 3, 5\}$; that is, $\{1, 3, 5\} \cup \{1, 2, 3\} = \{1, 2, 3, 5\}$.

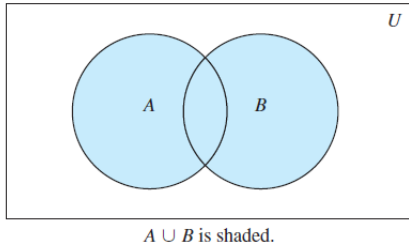


FIGURE 1 Venn Diagram of the Union of A and B .

Definition 12. Let A and B be sets. The intersection of the sets A and B , denoted by $A \cap B$, is the set containing those elements in both A and B .

An element x belongs to the intersection of the sets A and B if and only if x belongs to A and x belongs to B . This tells us that :

$$A \cap B = \{x \mid x \in A \wedge x \in B\}$$

. The intersection of the sets $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{1, 3\}$; that is, $\{1, 3, 5\} \cap \{1, 2, 3\} = \{1, 3\}$.

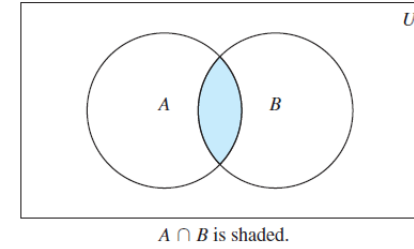


FIGURE 2 Venn Diagram of the Intersection of A and B .

$$\{1, 2, 3\} \cap \{1, 3, 5\} = \{1, 3\}.$$

Definition 13. Two sets are called disjoint if their intersection is the empty set.

Let $A = \{1, 3, 5, 7, 9\}$ and $B = \{2, 4, 6, 8, 10\}$. Because $A \cap B = \phi$, A and B are disjoint.

Remark : $|A \cup B| = |A| + |B| - |A \cap B|$.

Definition 14. Two sets are called disjoint if their intersection is the empty set.

Definition 15. Let A and B be sets. The difference of A and B , denoted by $A - B$, is the set containing those elements that are in A but not in B . The difference of A and B is also called the complement of B with respect to A .

An element x belongs to the difference of A and B if and only if $x \in A$ and $x \notin B$. This tells us that :

$$A - B = \{x \mid x \in A \wedge x \notin B\}$$

. The difference of $\{1, 3, 5\}$ and $\{1, 2, 3\}$ is the set $\{5\}$; that is, $\{1, 3, 5\} - \{1, 2, 3\} = \{5\}$. This is different from the difference of $\{1, 2, 3\}$ and $\{1, 3, 5\}$, which is the set $\{2\}$.

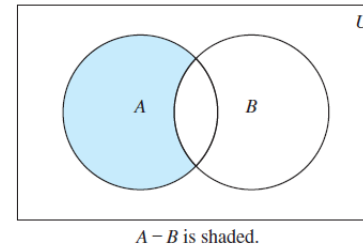
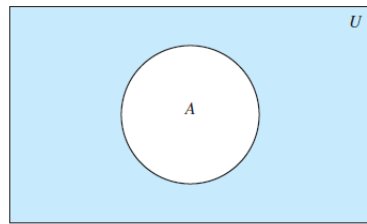


FIGURE 3 Venn Diagram for the Difference of A and B .

Definition 16. Let U be the universal set. The complement of the set A , denoted by $\bar{A}$, is the complement of A with respect to U . Therefore, the complement of the set A is $U - A$.

An element belongs to $\bar{A}$ if and only if $x \notin A$. This tells us that $\bar{A} = \{x \in U \mid x \notin A\}$.



$\bar{A}$ is shaded.

FIGURE 4 Venn Diagram for the Complement of the Set A .

Let $A = \{a, e, i, o, u\}$ (where the universal set is the set of letters of the English alphabet). $\bar{A} = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$.

Let A be the set of positive integers greater than 10 (with universal set the set of all positive integers). Then $\bar{A} = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.

3 Set Identities

TABLE 1 Set Identities.

Identity	Name
$A \cap U = A$ $A \cup \emptyset = A$	Identity laws
$A \cup U = U$ $A \cap \emptyset = \emptyset$	Domination laws
$A \cup A = A$ $A \cap A = A$	Idempotent laws
$\overline{(\bar{A})} = A$	Complementation law
$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
$A \cup (B \cap C) = (A \cup B) \cap C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
$\overline{A \cap B} = \bar{A} \cup \bar{B}$ $\overline{A \cup B} = \bar{A} \cap \bar{B}$	De Morgan's laws
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
$A \cup \bar{A} = U$ $A \cap \bar{A} = \emptyset$	Complement laws

Exercises :

1. Prove that $\overline{A \cap B} = \bar{A} \cup \bar{B}$.
2. Prove that $\overline{A \cup (B \cap C)} = (\bar{C} \cup \bar{B}) \cap \bar{A}$.
3. Prove that $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
4. Prove that $A \cap [(B \cup \bar{A}) - B] = \emptyset$

4 Exercises

Exercises 1 :

Let $U = \{x \in \mathbb{Z}^+ \mid -5 \leq x \leq 5\}$, $A = \{1, 2\}$, $B = \{1, 2, 5\}$, $C = \{1, 3, 5\}$ and $D = \{x \in U \mid x^2 - 36 = 0\}$.

Answer the following :

1. $(A \cap B) - D$
2. $\overline{(A \cap C) \cup B}$
3. $\overline{(B \cup C) - A \cup D}$
4. $\overline{(B - C) - A}$
5. $P(B - A) \cup A$
6. $P(\overline{D - (B - C)})$
7. $P(A) \cap P(B - C)$
8. $|P(A - B) \cup P(D - A)|$
9. $P(\overline{(A \cup D) \cup \emptyset})$.
10. $|P(P(P(A - \overline{D})))|$.
11. $|P(P(C - A) \cup (C \cup D))|$

Exercises 2 : Venn Diagram

Answer the following :

1. Shade $(A \cap B) - C$.
2. shade $\overline{((A \cap B) - C)}$.
3. Shade $\overline{A} - (C - B)$.
4. Given $A \subseteq B$ and $C - B = \emptyset$, shade $A \cap C = \emptyset$.
5. Shade the area for : $(A \cap \overline{B}) \cup (\overline{A} \cap C)$.
6. Shade $\overline{A - B} \cap \overline{B - C}$,
7. given $A \subseteq B$ and $B \cap C = \emptyset$, Shade $(A \cap B) \cup (\overline{(B \cup C)})$.
8. Given : $A \subseteq B$ and $|C - B| > 0$ and $A \cap C = \emptyset$ shade $(B - A) \cup C$.

Exercises 3 : Identities Proofs

1. Show $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$
2. Show that $(A \cap [(B \cup \overline{A}) - B]) = \emptyset$
3. Show that $(A - B) \cup (A - C) = A - (B \cap C)$.
4. Show that $\overline{A} \cup \overline{B} \cup (A \cap B \cap \overline{C}) = \overline{A} \cup \overline{B} \cup \overline{C}$
5. For any sets A, B , and C . Show that $\overline{(A \cup B \cup C)} \cup (A - (B \cup C)) = \overline{B \cup C}$.
6. Show using set identities that $(A - B) - (B - A) = (A - B)$.
7. Show using set identities that $(A - C) \cup (C - B) = \emptyset$.

Exercises 4 : MemberShip Proofs

1. if $A \subseteq B$ and $B \subseteq C$ then $A \subseteq C$
2. if $A \subseteq B$ then $A - B = \emptyset$
3. if $A - B = \emptyset$, then $A \subseteq B$
4. If $A \subseteq B$, then $B = A \cup B$
5. Show $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$ using membership proof.
6. show that $(C - D) \cup (C \cap D) \subseteq C$.
7. Prove by contradiction, for any sets A and B , if $A \subseteq A - B$, then $A \cap B = \emptyset$
8. For any set A, B, C if $A \subseteq B$ and $B \subseteq C$, then $(C - B) \cup A \subseteq C$.
9. Prove that if $A \cup C \subseteq B$, then $A \cap C \subseteq B$
10. Prove that For any set A, B, C if $A \subseteq B$ and $B \subseteq C$, then $B \cup A \subseteq C$.

Exercises 5 :

Examples – Counter Examples

1. For any set A, B, C : ff $A \subseteq B$ and $C \subseteq B$, then $|A| \cup |C| \leq |B|$
2. For any set A, C : $A \cap C = \emptyset \longrightarrow C - A = \emptyset$.
3. For any sets A and B , if $A \subseteq B$ and $|C - A| = 1$, then $|C - B| > 1$.
4. There are two sets A and B , $|A \cap B| = 0$ and $P(A - B) = \{\emptyset\}$
5. Given any set A, B . if $A - B = \emptyset$, then $B - A = \emptyset$.
6. For any sets A and B , if $A \subseteq B$, then $|A \cup B| = |B| + |A|$.
7. Given any set A, C : $A \cap C = \emptyset \longrightarrow C - A = \emptyset$.
8. For any sets A and B , if $A \subseteq B$ and $|C - A| = 1$, then $|C - B| > 1$.
9. there are two sets A and B , $|A \cap B| = 0$ and $P(A - B) = \{\emptyset\}$.
10. Given any set A, B : $A \cup B = B \longrightarrow B - A = \emptyset$.
11. For any sets A, B and C , if $A \subseteq C$ and $|A \cap B| = 0$, then $C \cap B = \emptyset$.
12. there are two sets A and B , $|P(A - B)| = 2$ and $P(B - A) = \{\emptyset\}$.